

Day 04 – Unit 4 : Growth Economics

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A. Growth vs Development vs Sustainable Development

1) Economic Growth

- Real national / **per capita income** mein long-run sustained increase.
- Nature → **Quantitative**, uni-dimensional, narrow. Measure: GDP, GNP, PCI.
- **Kuznets**: long-term rise in capacity to supply diverse goods.
- Output kitna bana ye batata hai — kis tak pahuncha (distribution) nahi batata.

2) Economic Development

- **Development = Growth + Structural & Distributional Change**
- Nature → Qualitative + Quantitative, multi-dimensional. Measure: HDI, PQLI, MPI.

Thinker	Core Idea
Dudley Seers	3 sawaal — Poverty, Unemployment, Inequality; teeno gire tabhi development
Todaro	3 core values — Sustenance, Self-esteem, Freedom
Amartya Sen	Expansion of capabilities & freedoms ("Development as Freedom")
Mahbub ul Haq	Human Development approach → HDI, HDR 1990
Schumpeter	Growth = gradual/continuous Development = discontinuous , innovation-driven
Clifford Geertz	"Growth without Development" (Indonesia study)

★ Growth is **necessary but NOT sufficient** for development. (Jobless / non-inclusive growth possible.)

3) Sustainable Development

- **Brundtland (1987)**, "**Our Common Future**": present needs meet karo bina future generations ki capacity compromise kiye. Commission ka asli naam = **WCED**.
- **3 Pillars**: Economic + Social + Environmental (People, Planet, Profit).
- **2 Equity**: **Inter-generational** (aaj vs kal) | **Intra-generational** (aaj ke logon mein).
- **Weak sustainability** = natural capital replaceable → **Hartwick Rule** (resource rents ko reproducible capital mein invest karo). **Strong** = irreplaceable.
- **Environmental Kuznets Curve** = inverted-U (income ↑ → pollution pehle ↑ phir ↓). Also: Green GDP.

📅 Milestones (ratification)

Year	Event
1972	Stockholm Conference + Limits to Growth (Club of Rome / Meadows)
1987	Brundtland Report — "Our Common Future"
1992	Rio Earth Summit → Agenda 21, UNFCCC, CBD
2000	MDGs — 8 goals (2000-2015)
2012	Rio+20 — "The Future We Want"
2015	SDGs — 17 Goals, 169 Targets, Agenda 2030 + Paris Agreement

B. Harrod-Domar Growth Model

Post-Keynesian • Harrod (1939) + Domar (1946) — alag-alag banaya, similar conclusion

Investment ka DUAL role —

- (i) **Demand effect** (Keynesian multiplier) → income badhta hai
- (ii) **Capacity effect** (Domar ka contribution) → productive capacity badhti hai
- ★ Steady growth ke liye dono ka balance zaroori.

Assumptions

- Initially full employment equilibrium; closed economy, no govt.
- Fixed capital-output ratio** — no substitution between K & L (Leontief).
- Saving ratio (s) constant; CRS; no depreciation; no technical progress; no lag.

Domar Model

$$\Delta I / I = \alpha \cdot \sigma$$

- α = MPS | σ = productivity of investment. Supply: $\Delta Y_s = I \sigma$; Demand: $\Delta Y_d = \Delta I (1/\alpha)$
- Conclusion:** sirf invest karna kaafi nahi — investment ka **growth rate** maintain karna zaroori hai.

Harrod — 3 Growth Rates ★★

Rate	Formula	Meaning
G — Actual	$G = s / C$	Jo growth actually realise hua (C = actual COR)
G_w — Warranted	$G_w = s / C_r$	Entrepreneurs satisfied , full capacity . "Entrepreneurial equilibrium" / self-warranting rate
G_n — Natural	labour growth + tech progress	Full employment CEILING . Harrod: "welfare optimum"

Instability I — Knife-Edge (Short-run): G vs G_w

Condition	Result
$G > G_w$	Actual COR < Required COR → capital shortage → aur investment → cumulative BOOM / inflation
$G < G_w$	Excess capacity → investment cut → cumulative DEPRESSION
$G = G_w$	Steady state — par <u>UNSTABLE</u>

- ★ Zara sa deviation aur economy dur bhagti jaati hai → "**Knife-edge / Razor's edge equilibrium**"

Instability II — Long-run: G_w vs G_n

- $G_n > G_w$ → chronic inflation, labour surplus, secular boom (developing countries).
- $G_n < G_w$ → chronic depression, **secular stagnation**, unemployment.
- $G_n = G_w = G$ → "**GOLDEN AGE**" (term by **Joan Robinson**).

$$g = s / v \rightarrow \text{Growth} \propto \text{saving rate} ; \text{Growth} \propto 1 / \text{capital-output ratio}$$

Criticism & India Link

- s aur v constant nahi; no factor substitution; technology & depreciation ignore; govt ka role nahi; developing countries pe applicable nahi (advanced capitalist economies ke liye bana).
- India: First Five Year Plan (1951-56) Harrod-Domar model pe based tha.**

C. Solow Neoclassical Growth Model (1956)

Solow + Swan, independently • Solow → Nobel 1987 • Purpose = knife-edge problem hatana

Assumptions

- $Y = F(K, L)$, Cobb-Douglas, **CRS**, **diminishing returns** to each factor.
- **K & L mein substitution possible** → **VARIABLE capital-output ratio** (yahi H-D se sabse bada farq).
- $S = sY = I$; labour force growth = n ; depreciation = δ ; technology **EXOGENOUS**.

Fundamental Solow Equation

$$\Delta k = s \cdot f(k) - (n + \delta)k$$

- $s \cdot f(k)$ = actual investment per worker.
- $(n + \delta)k$ = **break-even investment** → δk = depreciation replace; nk = **capital widening**.
- Jo bacha = **capital deepening** (per worker capital badhna).

Steady State ★★ $s \cdot f(k^*) = (n + \delta)k^*$

Variable	Growth rate at steady state
k (capital per worker), y (output per worker)	0
K (total capital), Y (total output)	n
Y with technical progress g	$n + g$
y with technical progress g	g ★

★ **Sabse bada conclusion:** Long-run per capita growth ka **ONLY source = technological progress** — saving rate se nahi!

Stability — Knife-edge SOLVED ✓

- $k < k^* \rightarrow s \cdot f(k) > (n + \delta)k \rightarrow k$ badhkar k^* pe aayega.
- $k > k^* \rightarrow s \cdot f(k) < (n + \delta)k \rightarrow k$ girkar k^* pe aayega. → **SELF-CORRECTING / STABLE**

Comparative Statics — Level Effect vs Growth Effect

Change	k^*, y^*	Long-run growth rate
Saving rate $s \uparrow$	Badhta $\uparrow$	NO change — sirf LEVEL effect ★
Population growth $n \uparrow$	Girta $\downarrow$	Y ka $\uparrow$; y ka 0 hi
Technology $g \uparrow$	—	Badhta — only true source
Depreciation $\delta \uparrow$	Girta $\downarrow$	No change

Golden Rule & Convergence

- **Golden Rule (Phelps, 1961):** per capita consumption maximum jahan **$MP_K = n + \delta$** . Agar $k^* > k_{gold} \rightarrow$ over-saving = **dynamic inefficiency**.
- **Absolute convergence:** poor countries tez grow karenge (k kam → MP_K zyada) = "catch-up effect".
- **Conditional convergence:** tabhi jab s, n, δ , technology similar ho — empirically ye zyada sahi. Bhi: Club convergence.

Growth Accounting / Solow Residual

$$\Delta Y/Y = \Delta A/A + \alpha(\Delta K/K) + (1-\alpha)(\Delta L/L)$$

- **$\Delta A/A$ = Solow Residual = TFP growth** — jo K, L se explain nahi hoti.
- **"Measure of our ignorance"** — phrase by Abramovitz.
- Solow ki US study (1909-49): output growth ka **~87.5% technical progress se**, sirf 12.5% capital se.

Criticism

- Technology **exogenous** ("manna from heaven") → isi gap se **Endogenous Growth Theory (Romer, Lucas)** aayi.
- Absolute convergence empirically fail; human capital ignore (→ Mankiw-Romer-Weil "Augmented Solow"); full employment unrealistic; Cambridge Capital Controversy (Joan Robinson).

D. MASTER TABLE — Harrod-Domar vs Solow ★★★

Basis	Harrod-Domar	Solow (Neoclassical)
School / Year	Post-Keynesian, 1939 / 1946	Neoclassical, 1956
Production function	Fixed proportion (Leontief)	Variable proportion (Cobb-Douglas)
Capital-output ratio	Fixed	Variable
Factor substitution	Not possible	Possible
Factor prices	Rigid	Flexible
Returns to capital	Constant	Diminishing
Equilibrium	UNSTABLE (knife-edge)	STABLE (self-correcting)
Full employment	Only by chance	Automatic (price flexibility)
Technology	Ignored	Exogenous
Long-run growth source	Saving rate (s)	Technical progress
Effect of $s \uparrow$	Permanent growth effect	Only level effect
Core equation	$G = s/C$; $\Delta I/I = \alpha \sigma$	$\Delta k = s \cdot f(k) - (n+\delta)k$
Nature	Short-run, disequilibrium	Long-run, equilibrium

E. Name ↔ Concept Match (Ratta List)

Concept	Name
Knife-edge equilibrium	Harrod
Capacity effect of investment	Domar
Golden Age	Joan Robinson
Golden Rule of capital accumulation	Phelps (1961)
Solow Residual / "measure of ignorance"	Abramovitz
Development as Freedom / Capability	Amartya Sen
HDI architect (HDR 1990)	Mahbub ul Haq
3 questions of development	Dudley Seers
Sustenance – Self-esteem – Freedom	Todaro
Growth vs Development distinction	Schumpeter
Sustainable Development definition	Brundtland / WCED
Hartwick Rule (weak sustainability)	Hartwick
Endogenous Growth Theory	Romer, Lucas
Augmented Solow (human capital)	Mankiw-Romer-Weil
Limits to Growth (1972)	Club of Rome / Meadows

F. Exam Traps — Yahin Galti Hoti Hai

- Solow mein s badhne se long-run growth rate **NAHI** badhta — sirf income ka level badhta hai. (#1 trap)
- $G > G_w \rightarrow$ **BOOM** (log ulta likh dete hai).
- Warranted** = entrepreneurs satisfied / full capacity. **Natural** = full employment ceiling.
- Golden Age** = Joan Robinson, **Golden Rule** = Phelps — mat mix karo.
- Steady state mein $k, y = 0$ growth, par $K, Y = n$ rate se grow karte hai.
- Knife-edge Solow ne **variable capital-output ratio** se solve kiya — yahi asli reason hai.
- Schumpeter: Growth = gradual, Development = discontinuous.
- Domar = σ (investment productivity) ; Harrod = C (capital-output ratio) — ye ek dusre ke **reciprocal** hai ($\sigma \approx 1/C$).
- Brundtland Commission ka asli naam = **WCED**. $SDG = 17$, $MDG = 8$.
- Solow-Swan — **Swan ka naam bhoolna nahi**.

✓ Day 04 Complete — Next: Day 05, Unit 5 (Money & Banking)